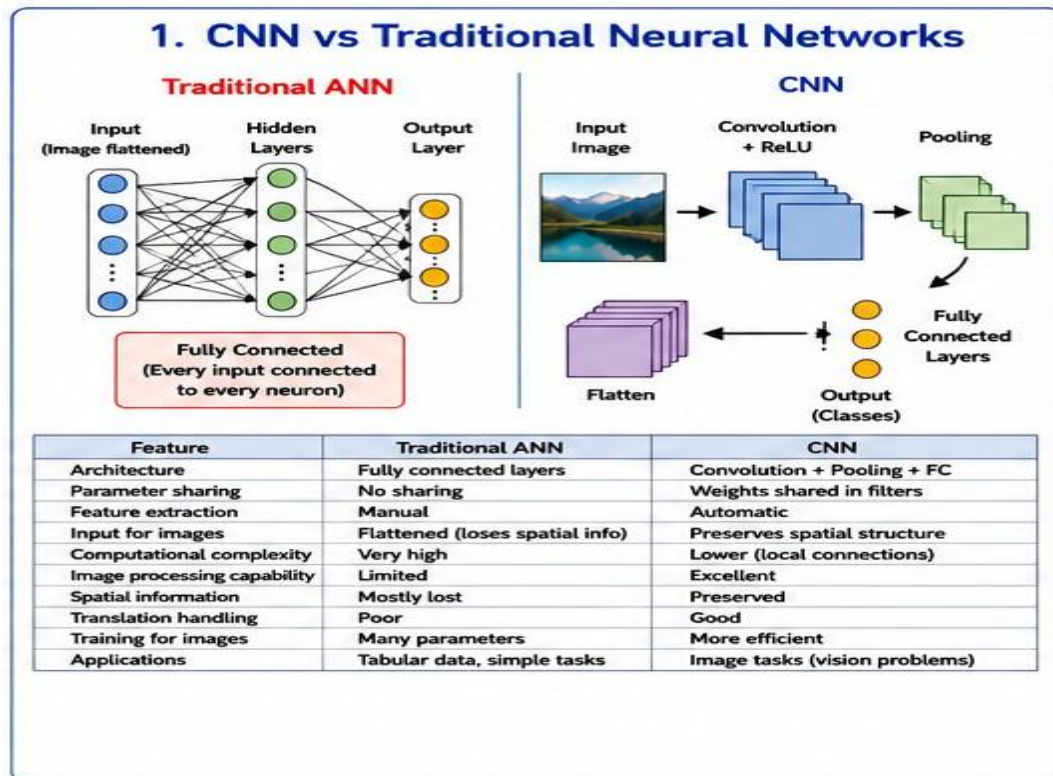


C02-AT2

M.POLIREDDY
REGNO:192325056
COURSE CODE:MLA0401
COURSE NAME:DEEP LEARNING

1. CNN vs Traditional ANN



Point ANN CNN

Architecture Fully connected Convolution + Pooling

Parameters More Fewer

Feature extraction Manual Automatic

Image processing Less suitable Highly suitable

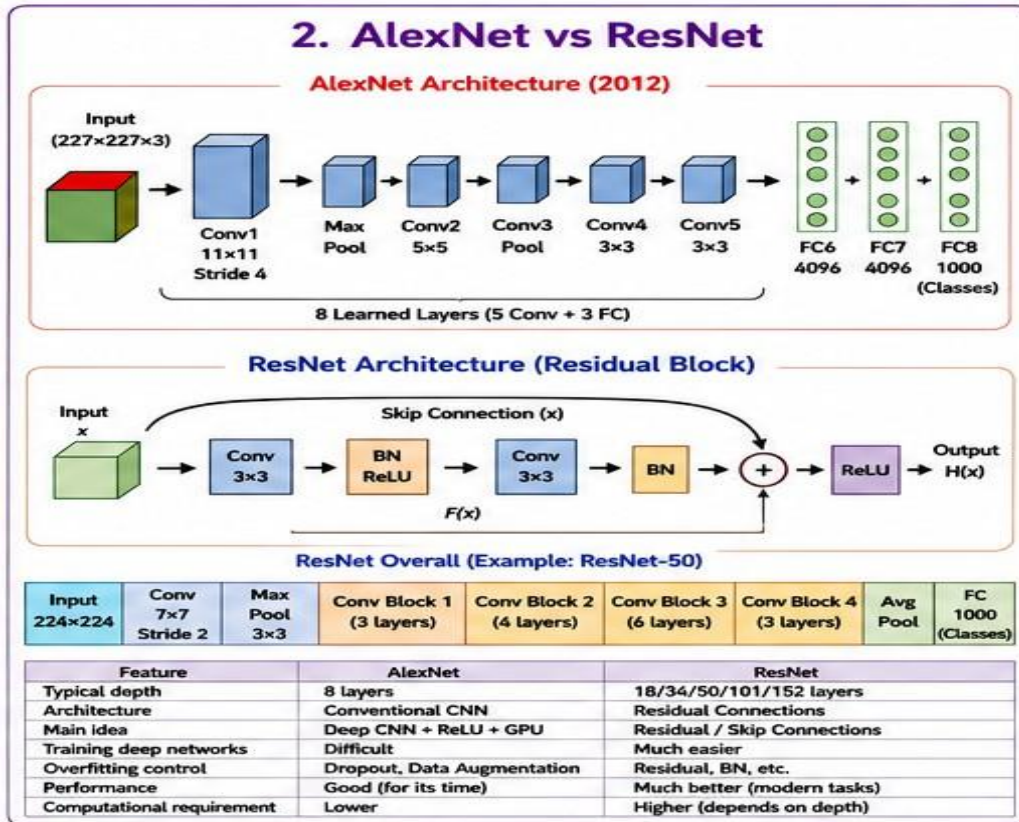
Spatial information Lost Preserved

Parameter sharing No Yes

Speed for images Slower Faster

Applications General data Images/videos

2. AlexNet vs ResNet



Point AlexNet ResNet

Year 2012 2015

Layers 8 18–152+

Architecture Normal CNN Residual CNN

Skip connection No Yes

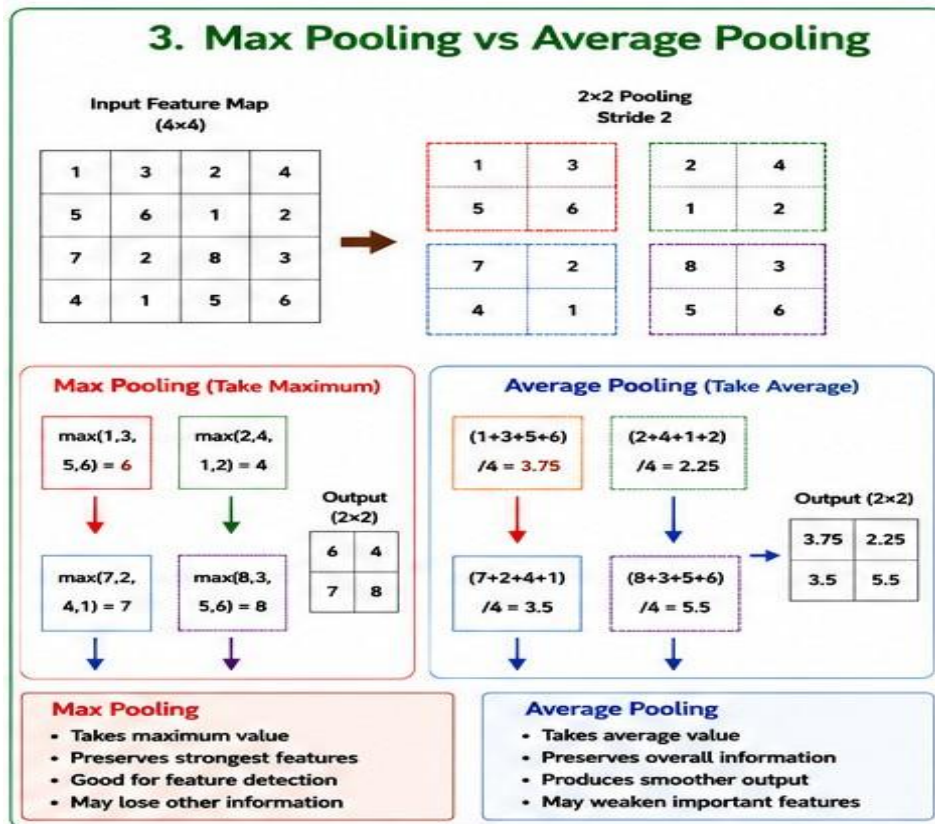
Training Difficult for deep networks Easier

Main feature ReLU + GPU Residual learning

Performance Good Better

Computation Lower Higher for deeper models

3. Max Pooling vs Average Pooling



Point Max Pooling Average Pooling

Operation Maximum value Average value

Feature Strongest feature Overall feature

Output Sharper Smoother

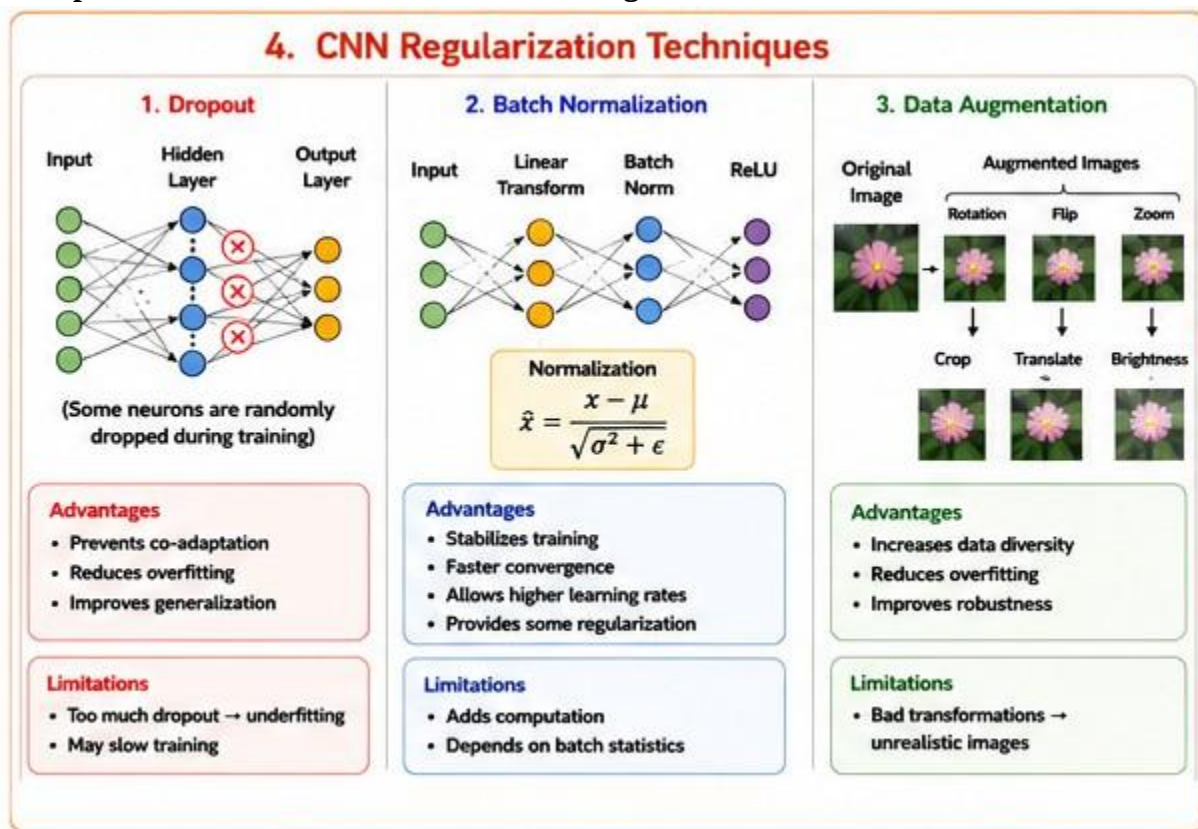
Advantage Keeps important features Keeps overall information

Disadvantage Loses some information May weaken features

Common use Image classification Feature smoothing

Example $\max(1,3,5,6) = 6$ $(1+3+5+6)/4 = 3.75$

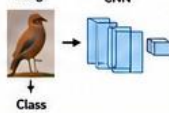
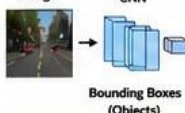
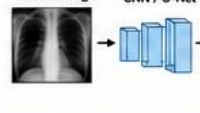
4. Dropout vs Batch Normalization vs Data Augmentation



Point	Dropout	Batch Normalization	Data Augmentation
Purpose	Reduce overfitting	Stable training	Increase data
Method	Removes neurons randomly	Normalizes activations	Modifies images
Example	Dropout = 0.5	Mean/variance normalization	Rotate, crop, zoom

Main benefit	Better generalization	Faster training	More diverse data
Limitation	Too much → underfitting	Needs batch statistics	Wrong changes can hurt
Applied to	Network	Network layers	Input images

5. CNN Applications

5. CNN Applications Analysis													
Application	Image Classification	Object Detection	Face Recognition	Medical Image Analysis	Autonomous Vehicles								
Main Task	Assign one class to an image	Locate and classify objects	Identify/verify a person	Detect / segment abnormalities	Understand road environment								
CNN Approach	 Image → CNN → Class	 Image → CNN → Bounding Boxes (Objects)	 Face Image → CNN → Face Embedding / Identity	 Medical Image → CNN / U-Net →	 Camera → CNN → Objects, Lanes, Road Understanding								
Example Output	<table><tr><td>Dog</td><td>0.05</td></tr><tr><td>Bird</td><td>0.90</td></tr><tr><td>Cat</td><td>0.03</td></tr><tr><td>Car</td><td>0.02</td></tr></table>	Dog	0.05	Bird	0.90	Cat	0.03	Car	0.02				
Dog	0.05												
Bird	0.90												
Cat	0.03												
Car	0.02												
Challenges	- Similar classes - Lighting, scale - Large datasets	- Multiple objects - Scale variation - Real-time processing	- Pose, lighting - Occlusion - Privacy	- Limited data - High accuracy - Interpretability	- Real-time - Weather, night - Safety critical								
Suitable CNN Models	ResNet, EfficientNet VGG, DenseNet	YOLO, SSD Faster R-CNN	FaceNet, ArcFace Siamese CNN	U-Net, ResNet 3D CNN	YOLO, SSD SegNet, DeepLab								

Application Purpose Example CNN

Image Classification Classify image ResNet

Object Detection Find + locate objects YOLO

Face Recognition Identify person Face CNN

Medical Analysis Detect diseases U-Net, ResNet

Autonomous Vehicles Detect road objects YOLO, SSD